

Transformers and MLflow: Downstream Tasks and Experiment Tracking

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Objective

Learn how to:

- Use pretrained transformer models for downstream tasks
- Fine-tune or run inference pipelines
- Track experiments, parameters, and results using MLflow
- Compare models systematically

You will implement three downstream tasks:

1. Text Classification (Sentiment Analysis)
2. Named Entity Recognition (NER)
3. Text Generation

Each task must be tracked using MLflow.

Task 1: Sentiment Classification

Goal: Use a pretrained model (e.g., `distilbert-base-uncased-finetuned-sst-2-english`)

Requirements:

- Use Hugging Face pipeline or Trainer API
- Run inference on at least 20 custom sentences
- Log the following in MLflow:
 - Model name
 - Number of samples
 - Accuracy
 - Predictions as an artifact (CSV)

Task 2: Named Entity Recognition

Goal: Use a pretrained NER model (e.g., `dslim/bert-base-NER`)

Requirements:

- Run on at least 10 paragraphs
- Extract entities:
 - Person
 - Organization
 - Location
- Log:
 - Model name
 - Total entities extracted
 - Distribution of entity types
 - Output a file as artifact

Task 3: Text Generation

Goal: Use a generative model (e.g., `gpt2`)

Requirements:

- Provide 5 prompts
- Generate text using different parameters:
 - temperature
 - max_length
- Log:
 - Parameters used
 - Generated outputs
 - Observations (as text artifact)

MLflow Requirements

For each task:

- Start an MLflow run
- Log:
 - Parameters (`mlflow.log_param`)
 - Metrics (`mlflow.log_metric`)
 - Artifacts (`mlflow.log_artifact`)
- Use meaningful experiment names:

Deliverables

1. Code (GitHub)
2. MLflow UI Screenshot (at least 2 runs per task)
3. A pdf of the Jupyter notebook with brief observations.